



**Piece
of charcoal**



**Diamond
crystal**

Allotropes

Several elements have multiple physical forms, or allotropes. For example, carbon has four allotropes: diamond, graphite, soot (or charcoal), and buckminsterfullerene. They are all made from pure carbon, but look and feel very different.

Radioactivity

While the main isotopes of most elements are stable, some rarer isotopes are less stable and the nuclei of their atoms break apart on their own. This process is called radioactivity, which releases dangerous particles and a lot of energy. These include positively charged alpha particles, negatively charged beta particles, and gamma rays, which carry no charge. Large, or heavy, atoms are less stable, and all elements with an atomic number above 82 are radioactive. Nuclear power stations use radioactivity from nuclear fission, a process in which atoms are split in two.

